

Amy Kang

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Experience

Software Engineering Intern | Meta, Datacenter Design & Config **May 2026 – Aug 2026**

- Engineered an automated regression safety net for Meta's datacenter switch config generation pipeline as part of the team's CI/CD workflow, replacing static-snapshot integration tests with dynamic baselines validated against production packages before comparison
- Automated the snapshot-update process and flagged risky diffs as "dangerous," increasing test signal fidelity and eliminating manual triage of false failures across the team's test suite
- Scaled the safety net across 13+ test networks, with continued expansion, covering configuration validation for approximately 500K devices

Software Engineering Intern | Amazon, Sustainable Shipping **May 2025 – Aug 2025**

- Delivered a sustainability-focused feature on Amazon's consumer platform, influencing millions of customer interactions
- Scaled A/B testing for 1M+ entries by re-engineering hard-coded configs into a DynamoDB database
- Built internal tooling enabling non-engineering teams to manage production configuration updates independently
- Optimized reads/writes to meet under-10ms latency requirements under production load

Software Intern (Machine Learning) | QueryStar, Analytics and Workflow Automation **May 2024 – Aug 2024**

- Trained a CNN-based game state classifier for racquet sports footage using PyTorch, improving model confidence by 20%
- Automated dataset labeling via Google Gemini Vision API, reducing manual annotation by 90%
- Enabled daily retrains by integrating iteration cycles and model validation into the production pipeline

Research & Projects

Elliptic Curve Group Structure Explorer | Directed Reading Program, Purdue Math **Jan 2026 – May 2026**

- Built and deployed a field-agnostic elliptic curve arithmetic playground (Python, Streamlit) for exploring the group structure of elliptic curves over arbitrary finite fields
- Implemented baby-step giant-step to compute curve group order in $O(\sqrt{n})$ time, then factorized results to determine cyclic group structure and generators

Research Assistant | Center for Computational and Applied Math, Purdue University **Nov 2024 – May 2025**

- Engineered MATLAB solvers using spherical harmonics to simulate elastic deformation of cells under microfluidic stress
- Formulated PDEs from elasticity and continuum mechanics
- Integrated models into numerical schemes, improving accuracy in boundary-driven strain simulations of elastic cells

Education

Purdue University, Main Campus | West Lafayette, IN **Aug 2023 – Expected May 2027**

B.S. in Computer Science & B.S. in Mathematics

GPA: 3.92

Relevant Coursework: Data Structures, Theory of Algorithms, Systems Programming, Computer Architecture, Operating Systems, AI, Probability

National University of Singapore | Study Abroad

Fall 2026

Coursework: Parallel Algorithms, Optimization Algorithms

Leadership & Activities

Teaching Assistant, CS 381 Analysis of Algorithms | Purdue CS Department **Aug 2025 – Present**

- Facilitated weekly recitations and office hours for a 500-student upper-level algorithms course

Hackathon Organizer | BoilerMake Hackathon, Purdue University **Apr 2024 – Present**

- Pitched selected event theme two years consecutively, shaping the branding and direction of a 500+ attendee hackathon; created sponsor materials and site UI in Figma and Affinity, contributing to \$80K+ in funding

Head Teaching Assistant, MA 366 Differential Equations | Purdue Math Department **Jan 2024 – May 2025**

- Directed MATLAB-based ODE lab sessions with 10 TAs and 5 faculty, redesigning problem sets for 120+ students

Technical Skills

Languages: Python, C, C++, Java, JavaScript, MATLAB, SQL

Frameworks & Libraries: Flask, PyTorch, OpenCV, Pandas, NumPy, Spring Boot, Flutter, Streamlit

Cloud & Tools: AWS (Lambda, DynamoDB, S3, CloudWatch), Google Cloud, Git, Hugging Face, CI/CD

Honors & Awards

1st Place, Popular Vote | Meta AI-VR Programming Hackathon, Menlo Park

Oct 2025

1st Place, Sustainability | InnovateHer Programming Hackathon, Purdue University

Feb 2024